

SITE / FACILITY INFORMATION			
FACILITY:	Cadillac WWTP	Digestion Method: 1 (Anaerobic = 1, Aerobic = 2, Lime stabilized = 3) App. Method: Injection (Inj, Inc, Sfc, not case sensitive) Org. N-1 yr. ago: Org. N-2 yrs. ago: TOTAL ACRES: 68 ACRES USED: 59 App. Rate (Gal/Ac) 16,136 App. Rate (DT/Ac) 2.30 Org. N App. (#/ac) 174	
MONTH & YEAR:	October 31, 2017		
FACILITY NPDES #	MI0020257		
MDEQ SITE ID #:	23N10W29-NS01		
COUNTY:	Wexford		
TOWNSHIP:	Colfax		
SECTION:	29		
OWNER:	Neil Smith		
FARMER:	Same		
ADDRESS:	1657 E. 24rd		
ADDRESS:	Cadillac, MI 49601		
TELEPHONE:	517-285-6349-		
SUPERINTENDANT	Jeff Dietlin		
LATITUDE:	N44* 21'55"		
LONGITUDE:	W85* 31'39		

Seasons Used:	3	Field Complete?	Yes
		Crop Year?	2018
		Overlap?	No

SOIL TEST RESULTS			
SAMPLE DATE:	08/28/17	NEXT CROP:	Corn Silage
CEC:	3.1	CROP YLD GOAL:	22 ba
PH:	5.9	NIT. REC. (lbs/ac):	220
LIME INDEX:	6.9	ADJ. N REC.	220
BRAY P1 (ppm):	63	AVAIL. N APP.	218
POTASSIUM (ppm):	51	SOIL & APP. P	320
CALCIUM (ppm):	294	P ₂ O ₅ REC. (lbs/ac):	0
MAGNESIUM (ppm):	35	K ₂ O REC. (lbs/ac):	215
		LIME REC. (T/ac):	2 ta

Accumulated History		
ARSENIC (As)	0.11	lbs/acre
CADMIUM (Cd)	0.04	lbs/acre
CHROMIUM (Cr)	0.47	lbs/acre
COPPER (Cu)	4.43	lbs/acre
LEAD(Pb)	0.25	lbs/acre
MERCURY (Hg)	0	lbs/acre
MOLYB. (Mo)	0.12	lbs/acre
NICKEL (Ni)	0.13	lbs/acre
SELENIUM (Se)	0.04	lbs/acre

[illegible]

ZINC (Zn)	6.72	lbs/acre
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	BIOSOLIDS ANALYSIS	
Lab ID	KAR	713302-01
Sample Name		Cadillac WWTP
Date Collected		9/12/2017
Matrix		Sludge
Density	lbs/gal	8.57
Mercury	mg/kg	0.57
Nitrogen,Ammonia	mg/kg	39,900.00
Nitrogen,Total	mg/kg	0.00
Nitrogen,Total Avail.	lb/ton	0.00
Nit.,Total Kjeldahl	mg/kg	77,900.00
pH	STD Units	7.90
Total Phosphorus	mg/kg	42,300.00
Nitrogen,Nitrate	mg/kg	16.00
Cadmium	mg/kg	3.80
Calcium	mg/kg	28,800.00
Chromium	mg/kg	89.50
Copper	mg/kg	825.00
Lead	mg/kg	38.00
Magnesium	mg/kg	4,900.00
Molybdenum	mg/kg	24.30
Nickel	mg/kg	26.60
Potassium	mg/kg	3,480.00
Zinc	mg/kg	1,250.00
Arsenic	mg/kg	22.00
Selenium	mg/kg	5.40
Total Solids	%	3.32
Total Volatile Solids	%	56.18

BIOSOLIDS APPLIED

[illegible]

Source:	Cadillac WWTP	Owner:	Neil Smith	App. Rate (Gal/Ac)	16,136
Date:	10/31/17	Farmer:	Same	App. Rate (DT/Ac)	2.30
NPDES #:	MI0020257	Address:	1657 E. 24rd	Useable Acres	68.0
MDEQ Site #:	23N10W29-NS01	City:	Cadillac, MI 49601	Acres Applied	59.0
County:	Wexford	Telephone:	517-285-6349-	Seasons Used	3
Township:	Colfax	Latitude:	N44* 21'55"		
Section:	29	Longitude:	W85* 31'39		

BIOSOLIDS ADDITIONS									
<u>Nutrient</u>	<u>Biosolids Additions</u>		<u>Residual Organic N Credit</u>		<u>Soil Fertility Test</u>			<u>Total Estimated Nutrients</u>	
Nitrogen (lb N/Ac)	218	+	0	+	0	lb N	=	218	lb N/Ac
Phos (lb P/Ac)	194			+	126	lb P	=	320	lb P/Ac
Pot (lb K/Ac)	16			+	102	lb K	=	118	lb K/Ac

SOIL ANALYSIS AND CROP INFORMATION					
C.E.C.:	3.1	P (lbs/acre):	126	K (lbs/acre):	102
Soil pH:	5.9	P (ppm):	63	K (ppm):	51
Lime Index:	6.9	Ca (lbs/acre):	588	Mg (lbs/acre):	70

FERTILIZER RECOMMENDATIONS (lbs./a)						
Crop to be Fertilized	Yield Goal	N	P2O5	K2O	Lime	
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Corn Silage	22 ba	220	0	215	2 ta	Lbs./A

BIOSOLIDS ANALYSES									
Date: 12-Sep		Dry wt. basis	Lbs/dry ton	Lbs Applied Per Acre	Allowable Lifetime Lbs/acre	Previous Applied lb/acre	Total Applied lb/acre	Life in Years	% Cum.
		-----	-----	-----	-----	-----	-----	-----	-----
Density	(mg/l)-->	1.03	----	----					
Weight	(Lb/Gal)>	8.57	----	----					
Solids	(%)----->	3.32	----	----					
TKN	(%)----->	7.79	----	----					
Amm. N	(%)----->	3.99	----	----					
Nit. N	(%)----->	0.002	----	----					
Total Plant Avail. N----->			95	218					
Total P	(%)--->	4.23	85	194					
Total K	(%)--->	0.35	7	16					
Total Ca	(%)--->	2.88	57.60	132.22					
Total Mg	(%)--->	0.49	9.80	22.50					
Total As	(ppm)->	22.00	0.044	0.101	37	0.11	0.21	366	0.57%
Total Cd	(ppm)->	3.80	0.008	0.017	35	0.04	0.06	2,006	0.16%
Total Cr	(ppm)->	89.50	0.179	0.411	N/A	0.47	0.88	N/A	N/A
Total Cu	(ppm)->	825.00	1.650	3.788	1,335	4.43	8.22	352	0.62%
Total Pb	(ppm)->	38.00	0.076	0.174	267	0.25	0.42	1,530	0.16%
Total Hg	(ppm)->	0.57	0.001	0.003	15	0.00	0.00	5,732	0.02%
Total Mo	(ppm)->	24.30	0.049	0.112	N/A	0.12	0.23	N/A	N/A
Total Ni	(ppm)->	26.60	0.053	0.122	274	0.13	0.25	2,244	0.09%
Total Se	(ppm)->	5.40	0.011	0.025	89	0.04	0.06	3,590	0.07%
Total Zn	(ppm)->	1250.00	2.500	5.739	2,492	6.72	12.46	434	0.50%

Month/Year: October-17		Michigan Department of Environmental Quality Water Division Biosolids Recycling Tracking Sheet Use of this form is voluntary. Submission of this form is not required														MDEQ Site # 23N10W29-NS01		CROP & SOIL DATA Crop to be fertilized: Corn Silage Subsequent Crop:													
Plant ID: MI0020257																# of seasons utilized: 3															
Superintendent: Jeff Dietlin		Acres utilized this month: 59																													
Facility Name: Cadillac WWTP		Total acres in site: 68																													
DAY	BIOSOLIDS APPLIED				NUTRIENTS								METALS																		
	MG	Solids %	Vol. %	Dry Tons	TKN %	NH4 %	NO3 %	TKN lb/ac	Phosphorus % lb/ac	Potassium % lb/ac	Arsenic mg/kg lb/ac	Cadmium mg/kg lb/ac	Copper mg/kg lb/ac	Lead mg/kg lb/ac	Mercury mg/kg lb/ac	Molybdenum mg/kg lb/ac	Nickel mg/kg lb/ac	Selenium mg/kg lb/ac	Zinc mg/kg lb/ac												
1	0.170	3.3		24.18454	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		Bray P1 (ppm): 63
2	0.204	3.3		29.02145	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		K (ppm): 51
3	0.187	3.3		26.60299	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		Crop Yield Goal: 22 ba
4	0.111	3.3		15.71995	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		Nitrogen Recom. (lb/ac) 220
5	0.153	3.3		21.76609	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
6	0.128	3.3		18.13841	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
7	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
8	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
9	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
10	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
11	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
12	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
13	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
14	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
15	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
16	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
17	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
18	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
19	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
20	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
21	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
22	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
23	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
24	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
25	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
26	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
27	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
28	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
29	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
30	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
31	0.000	0.0		0	7.79	3.99	0.0016		4.23		0.35		22.0		3.8		825		38.0		0.6		24.3		26.6		5.4		1250		
Average		3.32			7.79	3.99	0.0016		4.23		0.35		22.00		3.8		825		38.0		0.57		24.3		26.6		5.40		1250		
Totals	This Mo	0.952		135.4				218		194.2		16.0		0.1		0.0		3.8		0.2		0.0		0.1		0.1		0.0		5.7	BIOSOLIDS ANALYSIS DATE: 9/12/2017
	This Yr																													SOIL ANALYSIS DATE: 8/28/2017	
	Cumulative														0.21		0.06		8.22		0.42		0.00		0.23		0.25		0.06	12.46	

43039

Total acres in site: 68

K 102 lbs/a

[illegible]

[illegible]

Org. N-1 yr. Ago (lbs/ac):	0	0	Residual N Credit
Org. N-2 yrs. Ago (lbs/ac):	0	0	Residual N Credit

[illegible]

As	37
Cd	4.5
Cu	77.5
Pb	268
Hg	15
Ni	31
Se	89
Zn	155

[illegible]

26.60		5.40		1,250.00		2.88		0.49		Err:509
26.60		5.40		1,250.00		2.88		0.49		Err:509
26.60		5.40		1,250.00		2.88		0.49		Err:509
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26.60		5.40		1,250.00		2.88		0.49		Err:509
26.60		5.40		1,250.00		2.88		0.49		Err:509
26.60		5.40		1,250.00		2.88		0.49		Err:509
26.60		5.40		1,250.00		2.88		0.49		Err:509
26.60		5.40		1,250.00		2.88		0.49		Err:509
26.60		5.40		1,250.00		2.88		0.49		Err:509
0.122		0.025		5.739		132.220		22.496		Err:509
0.122		0.025		5.739		132.220		22.496		Err:509
0.137		0.028		6.427		148.086		25.195		Err:509
0.252		0.065		12.459		132.220		22.496		Err:509
0.282		0.073		13.954		148.086		25.195		Err:509

0.13	0.04	6.72	0	0	0
ni	se	zn	ca	mg	so4
1.286618	0.261193	60.46135	1393.0295	237.0085	Err:509
1.543941	0.313432	72.55362	1671.6354	284.4102	Err:509
1.415279	0.287312	66.50749	1532.33245	260.7093	Err:509
0.836301	0.169775	39.29988	905.469178	154.0555	Err:509
1.157956	0.235074	54.41522	1253.72655	213.3076	Err:509
0.964963	0.195895	45.34601	1044.77213	177.7564	Err:509
0	0	0	0	0	Err:509
0	0	0	0	0	Err:509
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0	0	0	0	0	Err:509
7.205058	1.462681	338.5836	7800.96522	1327.248	Err:509

Application Date	Facility	Facility #	Site I D	County	Township	Section
October 31, 2017	Cadillac WWTP	MI0020257	23N10W29-NS01	Wexford	Colfax	29

Owner	Farmer	Farmer Street Ad	Farm City, St. Zip	Frm Telephone	Entity Super	Latitutde
Neil Smith	Same	1657 E. 24rd	Cadillac, MI 49601	517-285-6349-	Jeff Dietlin	N44* 21'55"

Longitude	Season Used	Dig Method	App Method	Ttl Acres	Acres Used	Soil Date
W85* 31'39	3	1	Injection	68	59	8/28/2017

CEC	Ph	Lime Index	Bray P1	Pot	Calcium	Magnesium
3.1	5.9	6.9	63	51	294	35

Intend Crop	Yld Goal	N Rec	Phos Rec	Pot Rec	Lime Rec
Corn Silage	22 ba	220	0	215	2 ta

Facility	Facility #	Lab I.D.	Lab Ref #	Sample ID	Date Collected	Matrix	Density
Cadillac WWT	MI0020257	KAR	713302-01	Cadillac WW	9/12/2017	Sludge	8.57

Hg	N Amonia	N Total	N Ttl.AvaIL #/T	N Ttl Kjeldahl	pH	Ttl. Phos	N Nitrate
0.57	39900	0	0	77900	7.9	42300	16

Cd	Ca	Cr	Cu	Pb	Mag	Mo	Ni
3.8	28800	89.5	825	38	4900	24.3	26.6

K	Zn	As	Se	Ttl. Solids	Ttl. Vol. Solids
3480	1250	22	5.4	3.32	56.18

Facility	Facility #	Site ID	Date 1	Gallons 1	% Solids 1	Date 2	Gallons 2
adillac WW	MI0020257	3N10W29-NSC	10/31/2017	170,000	3.32	11/1/2017	204,000

% Solids 2	Date 3	Gallons 3	% Solids 3	Date 4	Gallons 4	% Solids 4	Date 5
3.32	11/2/2017	187,000	3.32	11/3/2017	110,500	3.32	11/6/2017

Gallons 5	% Solids 5	Date 6	Gallons 6	% Solids 6	Date 7	Gallons 7	% Solids 7
153,000	3.32	11/7/2017	127,500	3.32			

Date 8	Gallons 8	% Solids 8	Date 9	Gallons 9	% Solids 9	Date 10	Gallons 10
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% Solids 10	Date 11	Gallons 11	% Solids 11	Date 12	Gallons 12	% Solids 12	Date 13
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Gallons 13	% Solids 13	Date 14	Gallons 14	% Solids 14	Date 15	Gallons 15	% Solids 15
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Date 16	Gallons 16	% Solids 16	Date 17	Gallons 17	% Solids 17	Date 18	Gallons 18
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% Solids 18	Date 19	Gallons 19	% Solids 19	Date 20	Gallons 20	% Solids 20	Date 21
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Gallons 21	% Solids 21	Date 22	Gallons 22	% Solids 22	Date 23	Gallons 23	% Solids 23
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Date 24	Gallons 24	% Solids 24	Date 25	Gallons 25	% Solids 25	Date 26	Gallons 26
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% Solids 26	Date 27	Gallons 27	% Solids 27	Date 28	Gallons 28	% Solids 28	Date 29
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Gallons 29	% Solids 29	Date 30	Gallons 30	% Solids 30	Date 31	Gallons 31	% Solids 31
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